

Local Convolutions of Token Embeddings Improve Character-Level Discrete Diffusion Language Models

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Abstract

Discrete diffusion language models generate text by iteratively denoising a fully corrupted sequence in parallel, avoiding the sequential bottleneck of autoregressive decoding. We study a compact character-level discrete diffusion model (under 6.5M parameters) built on a DiT-style transformer and trained on the Tiny Shakespeare corpus under a fixed two-epoch budget on a single 16 GB GPU. Using an autonomous search of roughly 105 controlled experiments, we examine which architectural and optimization choices reduce validation loss in this constrained regime. Our main result is that applying a small stack of depthwise convolutions to the token embeddings, before the transformer, improves the model at very low cost: the two convolutions add about 2,300 parameters and a negligible amount of computation, yet they reduce validation loss by about the same percentage as changes that require many more parameters or much more compute, and they noticeably improve the quality of generated text. We also report the broader ablation, covering positional encoding, locality bias, noise-level conditioning, optimizer and schedule choices, and weight averaging, together with larger techniques (mixture-of-experts, linear attention, gated feed-forwards, sharpness-aware minimization) that did not help in this shallow, short-budget setting, and we close with a forward-pass kernel optimization that reaches a $1.81\times$ speedup.

1 Introduction

Autoregressive language models generate text one token at a time, which imposes three structural costs: generation is inherently $\mathcal{O}(N)$ in sequence length; the causal factorization makes it impossible to revise or inpaint earlier tokens without regenerating everything that follows; and decoding becomes memory-bound as the key-value cache grows with context length. Discrete diffusion language models (Austin et al., 2021; Lou et al., 2024; Sahoo et al.,

2024) offer an alternative. Starting from a fully corrupted sequence, the model iteratively denoises all positions in parallel, which enables bidirectional context, inpainting, and a compute/quality trade-off through the number of denoising steps. Recent work has scaled this paradigm to billions of parameters (Nie et al., 2025; Arriola et al., 2025), but the inductive biases that make small such models train well remain understudied.

This paper reports a systematic study of a tiny character-level discrete diffusion model. We adopt the nanoGPT (Karpathy, 2022) backbone, modernize it into a diffusion transformer (DiT) (Peebles and Xie, 2023) conditioned on the noise level through adaptive layer normalization (adaLN), and train it on Tiny Shakespeare at the character level under fixed hardware constraints (a single RTX 5080 Ti, 16 GB; at most 3 layers, 2 heads, embedding width 384, and batch size at most 512). Every configuration is evaluated by the same standardized two-epoch benchmark, so validation loss is directly comparable across experiments. The search was run as an autonomous loop: an agent proposes one testable hypothesis, implements it, runs the benchmark, and commits the change only if validation loss improves, reverting it otherwise.

Our central finding from roughly 105 logged experiments is that one of the most favourable changes is also one of the cheapest: a small stack of depthwise convolutions applied to the token embeddings before the transformer improves the model at very low parameter and compute cost, and also improves the qualitative output (Section 3). We additionally report a broad ablation over architecture and optimization, including techniques that did not help in this regime (Section 4), show generated samples (Section 5), and report a forward-pass kernel optimization (Section 6).

2 Model and Training Setup

We use an absorbing-state (masking) forward process in the style of MDLM (Sahoo et al., 2024) together with the score-entropy objective of SEDD (Lou et al., 2024). A clean sequence x_0 is corrupted at a continuous time $t \in [0, 1]$ under a cosine noise schedule, so the probability that a token is masked is $\bar{\sigma}_t = 1 - \cos(t\pi/2)$, rising from about 0 at $t=0$ to about 1 at $t=1$. The model $s_\theta(x_t, \bar{\sigma}_t)$ is trained to recover x_0 . We use two losses during the study: a time-conditioned masked cross-entropy on the corrupted positions, and the score-entropy objective, in which the rate-of-noise term σ_t weights the per-position loss,

$$\mathcal{L} = \mathbb{E}_{t, x_t} [\sigma_t \cdot \ell_{\text{SE}}(s_\theta(x_t, \bar{\sigma}_t), x_t, x_0)]. \quad (1)$$

The σ_t weighting is load-bearing for the score-entropy derivation: replacing it with uniform weighting strongly degraded results. Noise levels are drawn with antithetic sampling, pairing $(t, 1-t)$ within each batch.

The backbone is a pre-norm transformer with per-block adaLN modulation that derives shift, scale, and gate vectors from a sinusoidal embedding of the noise level (Peebles and Xie, 2023). Onto this backbone we layered the components found during the search (Section 4): rotary position embeddings (RoPE) (Su et al., 2024) with a learnable per-head ALiBi locality bias (Press et al., 2022); query-key normalization (Henry et al., 2020); a Dynamic Tanh normalization (Zhu et al., 2025) as a faster replacement for LayerNorm; and two zero-initialized projections that inject the noise level directly at the embedding and logit levels. The model stays under 6.5M parameters, and sampling uses MaskGIT-style confidence-based iterative unmasking (Chang et al., 2022). Figure 1 shows the full architecture, including the embedding convolution studied next.

3 Convolutional Preprocessing of Token Embeddings

A standard transformer block mixes information across positions only through self-attention, so at the character level the model must learn even the most local structure, such as common digraphs, morphemes, and word boundaries, from attention alone. We instead give the model an explicit local operator by convolving the token embeddings before they enter the transformer. After the embedding lookup produces h , we apply a residual

Addition	Val. red.	Params	Compute
Embedding conv stack	0.9%	2,304	negligible
Register tokens (8)	0.6%	3,072	higher
Context 384→512	0.5%	0	higher

Table 1: Validation-loss reduction (relative) versus cost for three additions in the cross-entropy ablation. The embedding convolution gives the largest reduction of the three while adding few parameters and almost no compute, whereas register tokens add more parameters and a longer attended sequence, and a longer context window raises attention cost quadratically.

stack of two depthwise (groups = n_{embd}) one-dimensional convolutions with kernel size 3 and a nonlinearity before the second:

$$h \leftarrow h + \text{DWConv}_3(h), \quad (2)$$

$$h \leftarrow h + \text{DWConv}_3(\text{SiLU}(h)). \quad (3)$$

The two convolutions give an effective receptive field of five characters with a nonlinear interaction between them, and both are residual, so the stack starts near the identity and gradually specializes into local feature detectors. This is the only convolutional component in the final model; it acts once, on the embeddings, and not inside the transformer blocks.

The addition is inexpensive. A depthwise convolution with kernel size k over an embedding of width n_{embd} has $n_{\text{embd}} \cdot k$ weights, so the two layers add $2 \cdot 384 \cdot 3 = 2,304$ parameters, under 0.04% of the budget. Its forward cost is $\mathcal{O}(n_{\text{embd}} k T)$ for a length- T sequence, negligible next to the $\mathcal{O}(T^2 n_{\text{embd}})$ attention and $\mathcal{O}(T n_{\text{embd}}^2)$ feed-forward cost of even one block.

In the controlled ablation, introducing the convolution on the embeddings reduced validation loss by about 0.5%, and deepening it into the two-layer stack of Equations 2–3 reduced it by a further 0.4%, for a combined reduction of roughly 0.9%. Table 1 places this against two other changes of similar magnitude. The convolution achieves the largest of the three reductions while adding fewer parameters than the register tokens and far less compute than the longer context window, making it one of the most cost-effective changes in the study.

The benefit was not only quantitative. Of all the changes we made, the embedding convolution produced the clearest improvement in sample quality: the pre-convolution model tended to emit degenerate repetitions such as “your ... your ... your”, whereas the convolutional model produced

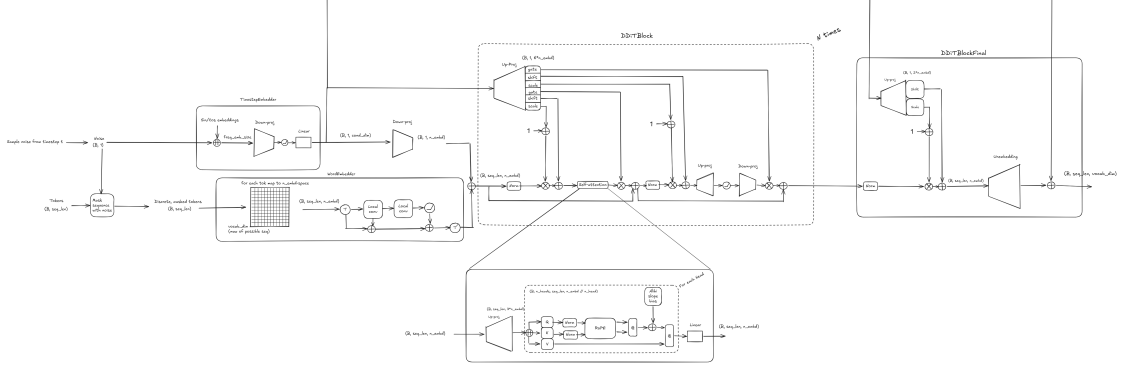


Figure 1: The diffusion transformer backbone. A scalar noise level σ is embedded and used both for per-block adaLN modulation (shift, scale, gate) and for direct embedding-level and logit-level bypass projections. Token embeddings pass through a stack of depthwise convolutions (left, “local conv”) before entering the transformer blocks (Section 3).

well-formed words, plausible character names, and sentence-like structure. This is consistent with the mechanism, since a depthwise convolution is a direct inductive bias for the short-range, position-local correlations (spelling and common n -grams) that dominate character-level text and that parallel, order-agnostic denoising otherwise has to recover through attention. The receptive field mattered: kernel size 3 (receptive field 5 when stacked) outperformed kernel size 5 and dilated variants, indicating that tight locality is what helps. The recipe is also specific: normalizing before each convolution, applying it before rather than after the residual embedding, zero-initializing the convolutions, and adding a convolution on the final pre-logit representation all regressed. Convolutions inside every transformer block helped under the cross-entropy proxy but gave no reliable benefit under score-entropy, so the final model keeps only the single embedding-level stack.

4 Ablation Study

Each experiment tests one hypothesis under the same two-epoch benchmark; a change is kept only if it lowers validation loss. This greedy procedure over about 105 experiments produced the cumulative milestones in Table 2; most candidate changes landed at or above the running best, showing how few actually help in this regime.

Beyond the embedding convolution, the largest single improvement was RoPE (Su et al., 2024), which replaced learned positional embeddings and reduced validation loss by about 4.4%, with smaller additional gains from the learnable per-head ALiBi locality bias (Press et al., 2022) and query-key

Milestone	Val. loss	Cum. red.
Baseline (default config)	1.0268	—
+ timestep (σ) embedding	1.0105	1.6%
+ RoPE	0.9152	10.9%
+ embedding convolutions	0.8832	14.0%
+ ALiBi and QK-Norm	0.8770	14.6%
+ cosine LR with warm restarts	0.8531	16.9%
+ weight averaging (decay 0.998)	0.8203	20.1%
+ batch-size tuning	0.7960	22.5%
Final	0.7920	22.9%

Table 2: Cumulative milestones of the autonomous search on the cross-entropy ablation path, with reduction relative to the baseline. The convolution row aggregates input and per-block experiments; the final model retains only the input stack (Section 3).

normalization (Henry et al., 2020). Direct noise-level conditioning, obtained by scaling the noise level before the sinusoidal embedding and adding zero-initialized projections of it at the embedding and logit levels, gave consistent small gains. On the optimization side, the Muon optimizer (Jordan et al., 2024) applied to the two-dimensional weight matrices, with AdamW for the rest, tolerated a learning rate roughly twice that of AdamW; a cosine schedule with warm restarts (Loshchilov and Hutter, 2017) at one cycle per epoch, together with exponential moving averaging of the weights for evaluation, contributed the largest optimization gains. Smaller batch sizes (around 96) also outperformed larger ones, as more frequent updates and higher gradient noise improved generalization within the fixed budget.

Many plausible modern techniques regressed. Mixture-of-experts (Shazeer et al., 2017), gated linear-attention layers, and SwiGLU feed-forwards (Shazeer, 2020) all hurt: in a 3-layer, two-epoch

model the gating and routing mechanisms never get enough steps to specialize, and any reduction in dense feed-forward capacity is not recovered. Sharpness-aware minimization (Foret et al., 2021) doubled compute and still made validation worse; stochastic weight averaging (Izmailov et al., 2018) underperformed the exponential moving average, because the model is still improving monotonically, so any equal-weight window over snapshots includes worse early ones; and span masking broke the per-token independence assumed by the score-entropy derivation. A consistent lesson is that the model sits in a deep local optimum for the budget, and that weight averaging, rather than any single architectural trick, is the dominant source of final-model generalization. The milestone path used the masked cross-entropy loss, which yields lower numbers than score-entropy (whose final loss is about 0.82 with weight averaging); the ranking of changes, and the convolution result in particular, was consistent across both.

5 Generated Samples

Figure 2 shows a sample from the final model (averaged weights, 256 unmasking steps) from a fully masked sequence. A model of this size and budget does not produce semantically coherent passages, but the sample reproduces salient features of the corpus: speaker labels in capitals followed by a colon (MENENIUS:) appear in the correct position; the vocabulary and morphology are archaic and largely well-formed (*your honour, crook'd upon my gage*); and the line lengths and punctuation follow the dramatic-verse layout of the source. The errors are mostly local (*thae* for *the*, *mame* for *name*), as expected from a character-level model that has captured local orthographic and structural regularities but not long-range meaning. This matches the quantitative picture, since the embedding convolution improves exactly this kind of local structure and was also the change that most visibly improved the samples.

6 Forward-Pass Cost

Because diffusion sampling repeats the forward pass once per denoising step, forward-pass latency directly determines generation cost. We optimized the forward pass of the 6.47M-parameter model at the kernel level. Compilation with `torch.compile` gave a $1.58\times$ speedup, from 32.9 ms to 20.8 ms, and replacing the hottest opera-

to take your honour to you, my lord.

```
MENENIUS:
I will not come to have them, for you
  have you need,
To give me with your honour to your
  consent,
And you are your honour and your mother,
Your honour save you yet thae more,
You have done to the matters of the
  cause;
But the time shall be crook'd upon my
  gage,
And make my hearted the house of the
  nose,
That he is no more that mame
```

Figure 2: A sample from the final model. Speaker labels, archaic vocabulary, and verse layout are reproduced; the remaining errors are local. Long lines are wrapped to fit the column.

tions with custom Triton kernels (Tillet et al., 2019) reached $1.81\times$, at 18.2 ms. Since unmasking uses 256 steps, these constant-factor gains compound directly into end-to-end sampling throughput.

7 Conclusion

In a tightly constrained character-level discrete diffusion setting, one of the most effective and most transferable additions was also among the cheapest: a small residual stack of depthwise convolutions on the token embeddings before the transformer. It reduced validation loss by about the same percentage as changes costing many more parameters or much more compute, and it gave the clearest improvement in qualitative output, while adding only about 2,300 parameters and negligible computation, even as heavier techniques such as mixture-of-experts, linear attention, and sharpness-aware minimization regressed. We read this as evidence that, for character-level diffusion, an explicit local operator complements the global mixing of attention in a way that is otherwise hard for a small model to learn.

Limitations

The study is confined to one small dataset (Tiny Shakespeare) at the character level, one model under 6.5M parameters, a fixed two-epoch budget, and largely a single seed, so absolute numbers and the sign of some effects may differ at scale, on subword tokenization, or with longer training. The autonomous search is a greedy hill-climb that commits on small validation-loss differences, some

within run-to-run noise, so the path is not guaranteed optimal. Finally, validation loss is an imperfect proxy for generation quality, and the qualitative claims rest on sample inspection rather than human evaluation.

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